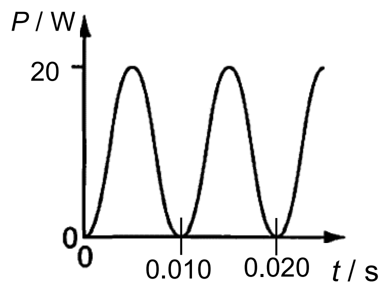
**27** A sinusoidal alternating current $I$ / A varies with time $t$ / s according to the equation

$$I = 2.0 \sin (100\pi t)$$

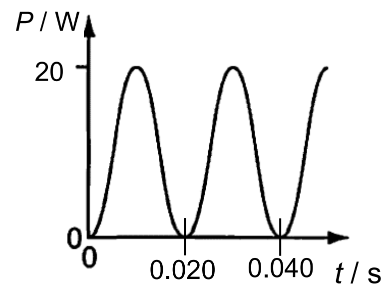
This current is allowed to pass through a resistor of resistance 5.0 Ω.

Which of the following shows how the power $P$ / W dissipated in the resistor varies with $t$ / s?

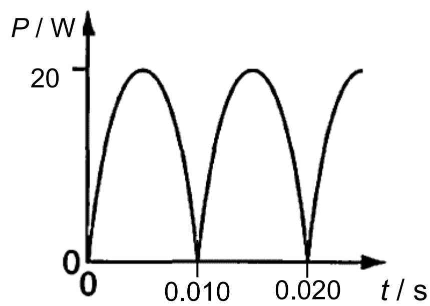
**A**



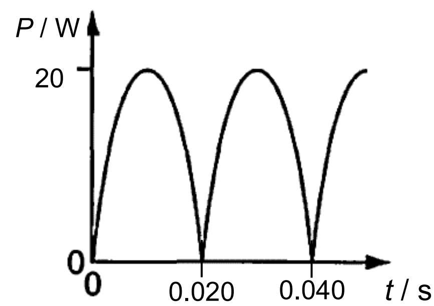

**B**




**C**




**D**




**Answer: A**

**From the equation, compare with the general equation: $I = I_o \sin (\omega t)$**
**$I_o = 2.0$ A      and      $\omega = 100\pi$**

**$P_o = I_o^2 R = (2.0^2)(5.0) = 20$ W**

**$T = 2\pi/\omega = 2\pi/100\pi = 0.020$ s**

**P-t graph is a sine-*squared* function.**

| | |
|---|---|
| 28 | The primary coil of an ideal transformer has 1000 turns and is connected to a sinusoidal |